LR-FHSS Power Model

Clear Theory Slides

What We Model

- Satellite LR-FHSS receiver
- Limited demodulators
- Battery + solar power
- Load-dependent decoding
- Header-only decode KPI (primary)
- Header-including-payload KPI (secondary)

State:

- N: node load
- D_{req}: requested demods
- B_t: battery SoC
- V: visibility (0/1)

Code Trace (Quick)

- N -> `nodes`
- D_{req} -> `requested_demods`
- D -> `allocated_demods`
- m_t -> `p_mode`
- V -> `visible`
- T -> `tx_count`
- H_{hdr} -> `decoded_headers`
- H_{inc} -> `decoded_headers_including_payloads`
- P_{cons} -> `power_consumption`
- P_{net} -> `net_power_watts`
- B_{t+1} -> `updated_battery_percent`

Why We Need This

- Decode performance alone is not enough.
- Energy limits change operating mode.
- We need decode + power in one loop.

Step 1: Mode Policy

$$m_t = \begin{cases} \text{sleep}, & B_t \leq B_{\text{low}} \text{ or } V = 0 \text{ or } N = 0 \text{ or } D = 0 \\ \text{idle}, & B_t < B_{\text{idle}} \\ \text{idle}, & N < 200 \text{ and } D \leq D_{\text{hi}} \\ \text{busy}, & \text{otherwise} \end{cases}$$

Variables: m_t operating mode, B_t battery SoC (%), B_{low} low-battery threshold, B_{idle} idle-mode threshold, V visibility flag, N node load, D allocated demods, D_{hi} high-charge demod threshold.

Step 2: Demod Allocation

$$D = \begin{cases} 0, & m_t = \text{sleep} \\ D_{\text{req}}, & \text{idle/busy} \end{cases}$$

Label: Demod allocation rule.

Variables: D allocated demods, D_{req} requested demods, m_t operating mode.

Step 3: Utilization

$$u = \min\left(1, \frac{T}{kD}\right), \quad u \in [0, 1]$$

Label: Utilization equation.
Variables: u utilization, T traffic demand per step, k per-demod step capacity, D allocated demods.

- T: traffic demand per step
- kD: decode capacity per step

Step 4: Power Consumption

$$P_{\text{cons}} = \begin{cases} P_0, & \text{sleep} \\ P_0 + D(a_i + b_i u), & \text{idle} \\ P_0 + D(a_b + b_b u), & \text{busy} \end{cases}$$

Variables: P_{cons} receiver power draw, P_0 baseline platform power, D allocated demods, u utilization, (a_i, b_i) idle coefficients, (a_b, b_b) busy coefficients.

Step 5: Solar Generation

$$P_{\text{gen}} = V P_{\text{solar}} \eta_{\text{pc}}$$

Label: Solar generation model.
Variables: P_{gen} generated power, V visibility flag, P_{solar} panel rating, η_{pc} power-conditioning efficiency.

Surplus:

$$S = P_{\text{gen}} - P_{\text{cons}}$$

Label: Power surplus.
Variables: S surplus power, P_{gen} generated power, P_{cons} consumption power.

Step 6: Net Battery Power

Charge taper:

$$\alpha(B_t) \in \{1, 0.5, 0.25, 0\}$$

Variables: $\alpha(B_t)$ charge-acceptance scale, B_t battery SoC.

$$P_{\text{ch}} = \min(\max(S, 0), P_{\text{ch,max}}\alpha(B_t))$$

Variables: P_{ch} charging power, S surplus power, $P_{\text{ch,max}}$ max charge power, $\alpha(B_t)$ charge-acceptance scale, B_t battery SoC.

$$P_{\text{dis}} = \max(-S, 0), \quad P_{\text{net}} = P_{\text{ch}} - P_{\text{dis}}$$

Variables: P_{dis} discharging power, S surplus power, P_{net} net battery power, P_{ch} charging power.

Step 7: Battery Update

$$\Delta E = \eta_{\mathrm{ch}} \max(P_{\mathrm{net}}, 0)\Delta t - \frac{1}{\eta_{\mathrm{dis}}} \max(-P_{\mathrm{net}}, 0)\Delta t$$

Variables: ΔE battery energy change, P_{net} net battery power, η_{ch} charging efficiency, η_{dis} discharging efficiency, Δt step duration.

$$B_{t+1} = \mathrm{clip}_{[0,100]}\left(B_t + 100\frac{\Delta E}{C_{\mathrm{Wh}}}\right)$$

Variables: B_{t+1} next-step SoC, B_t current SoC, ΔE battery energy change, C_{Wh} battery capacity (Wh), $\mathrm{clip}_{[0,100]}$ bound operator.

Full Combined Chain

$$(N, D_{\text{req}}, V, B_t) \to m_t \to D \to u \to P_{\text{cons}} \to P_{\text{net}} \to B_{t+1}$$

Label: Full simulation pipeline.
Variables: N node load, D_{req} requested demods, V visibility, B_t current SoC, m_t mode, D allocated demods, u utilization, P_{cons} consumption power, P_{net} net battery power, B_{t+1} next-step SoC.

Implementation note:

- In current code, battery is reset per scenario: $B_t = B_{\text{init}}$ for each (N, D_{req}) point.

When This Model Is Valid

- Early-stage mission trade studies
- Control/policy comparison
- Fast sweeps over load and demods

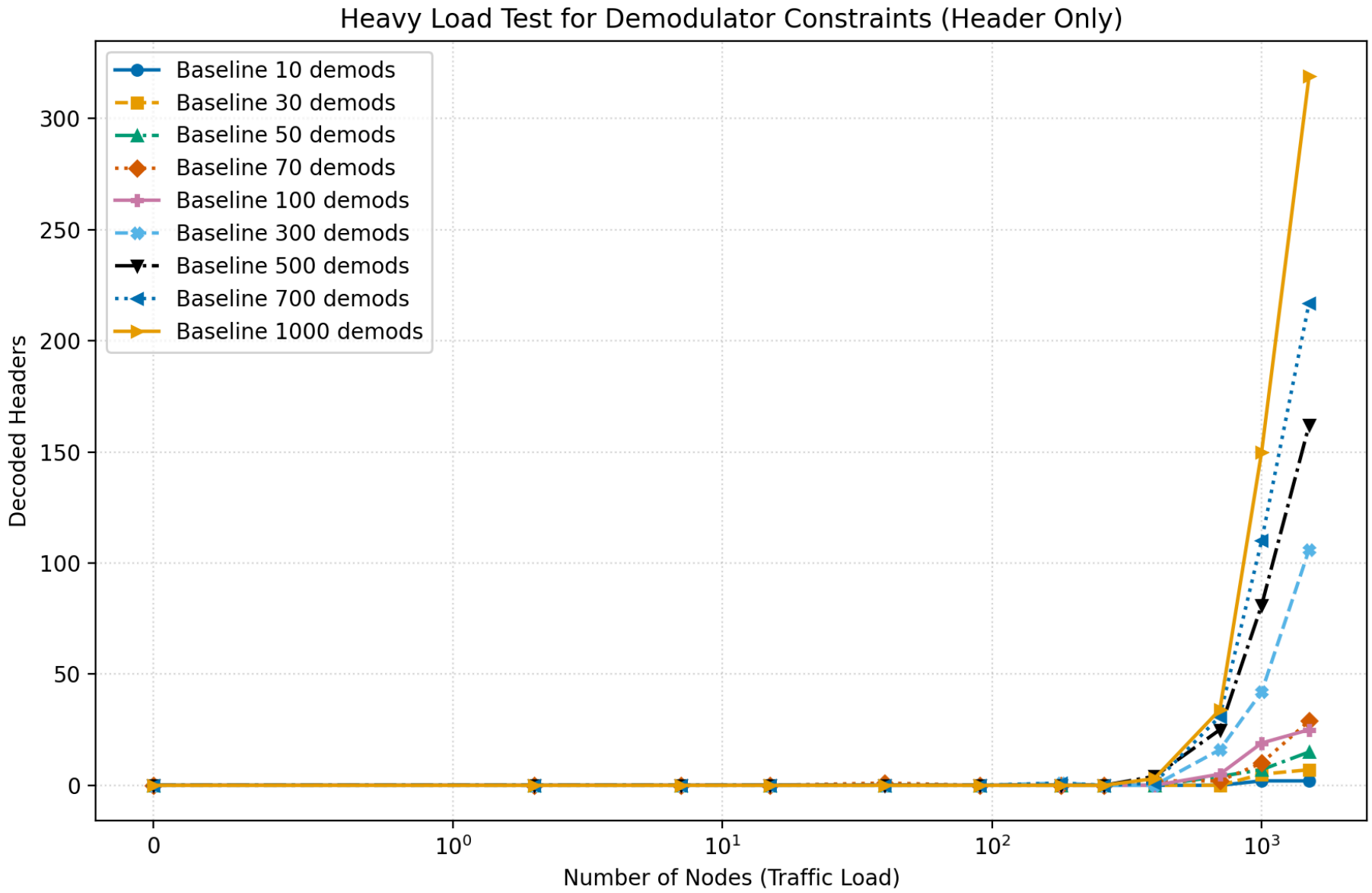
Not enough for:

- high-fidelity electrochemistry
- thermal-coupled battery physics

Results

Header-Only KPI

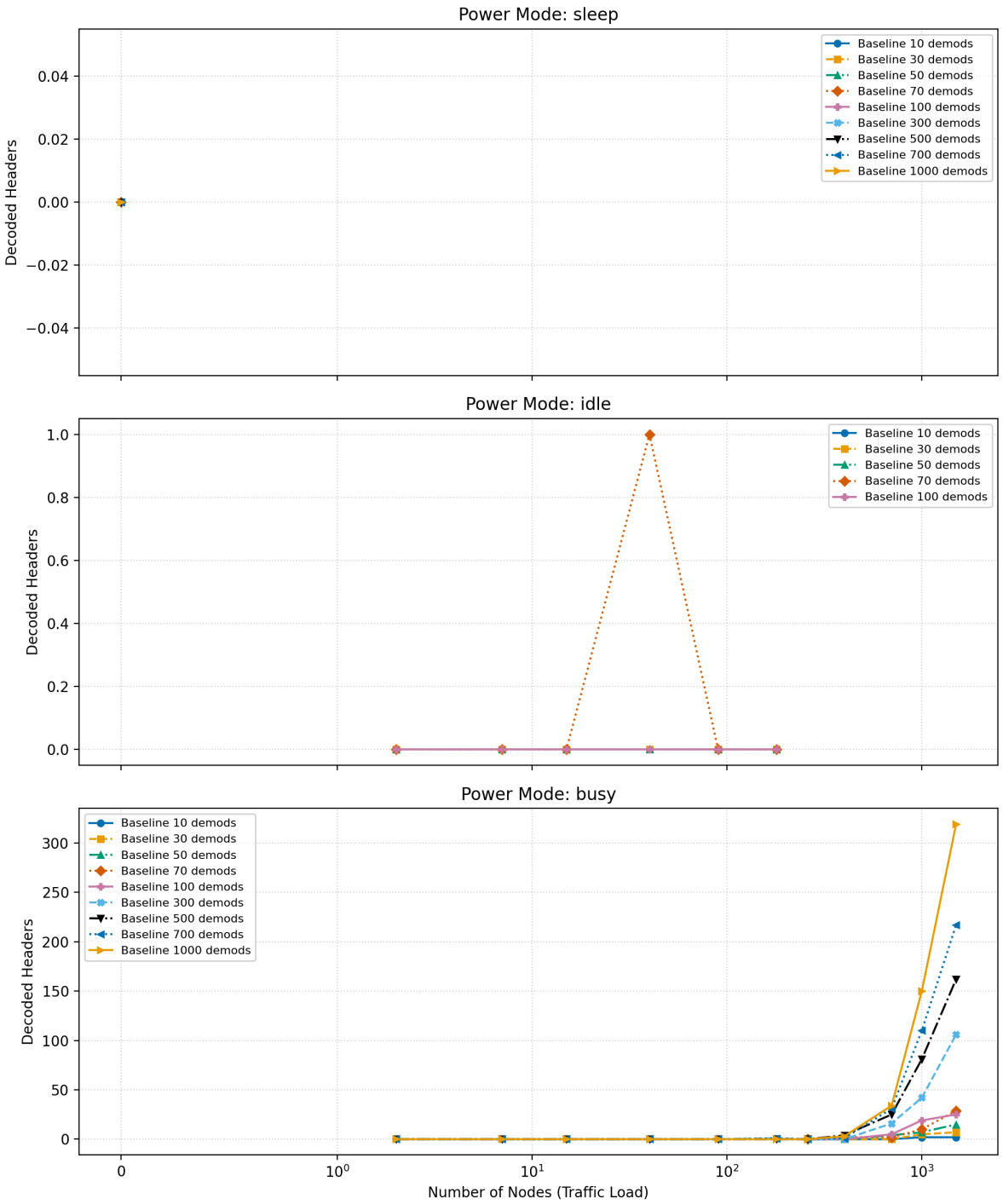
Primary metric for performance comparison.



Results

Header-Only by Mode

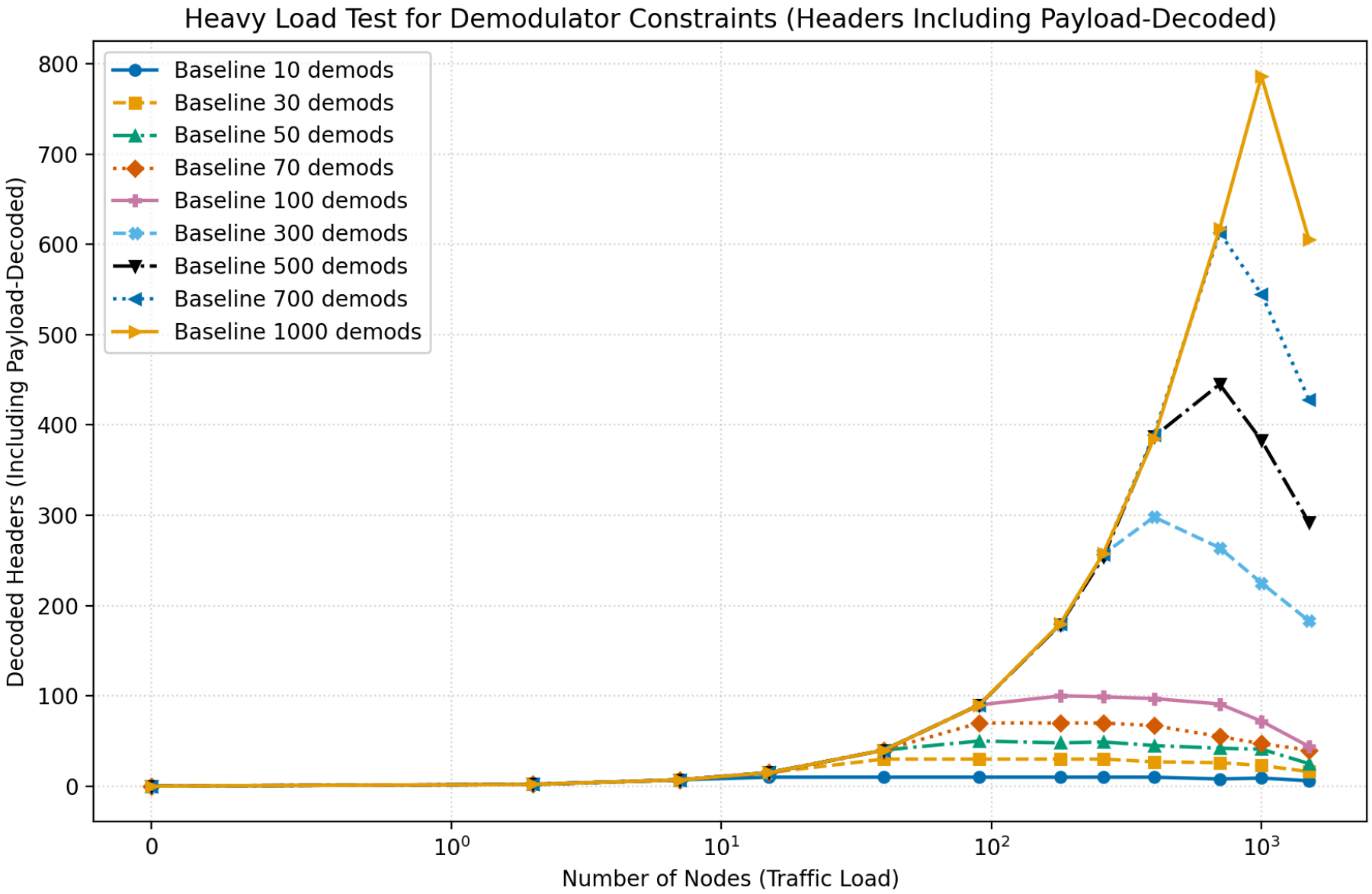
Sleep, idle, and busy behavior split.



Results

Header-Including-Payload KPI

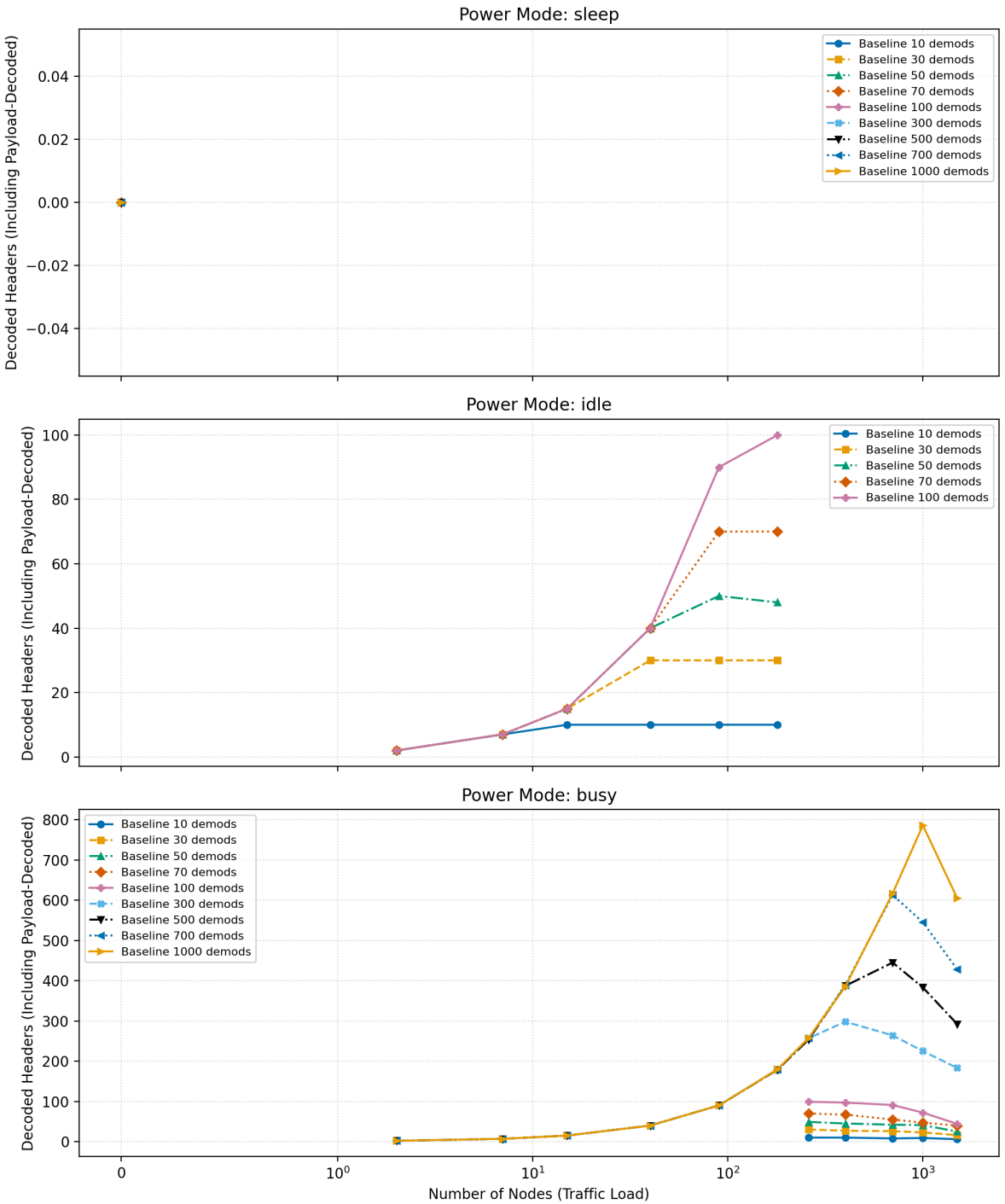
Secondary metric: header decode including payload-decoded packets.



Results

Header-Including-Payload by Mode

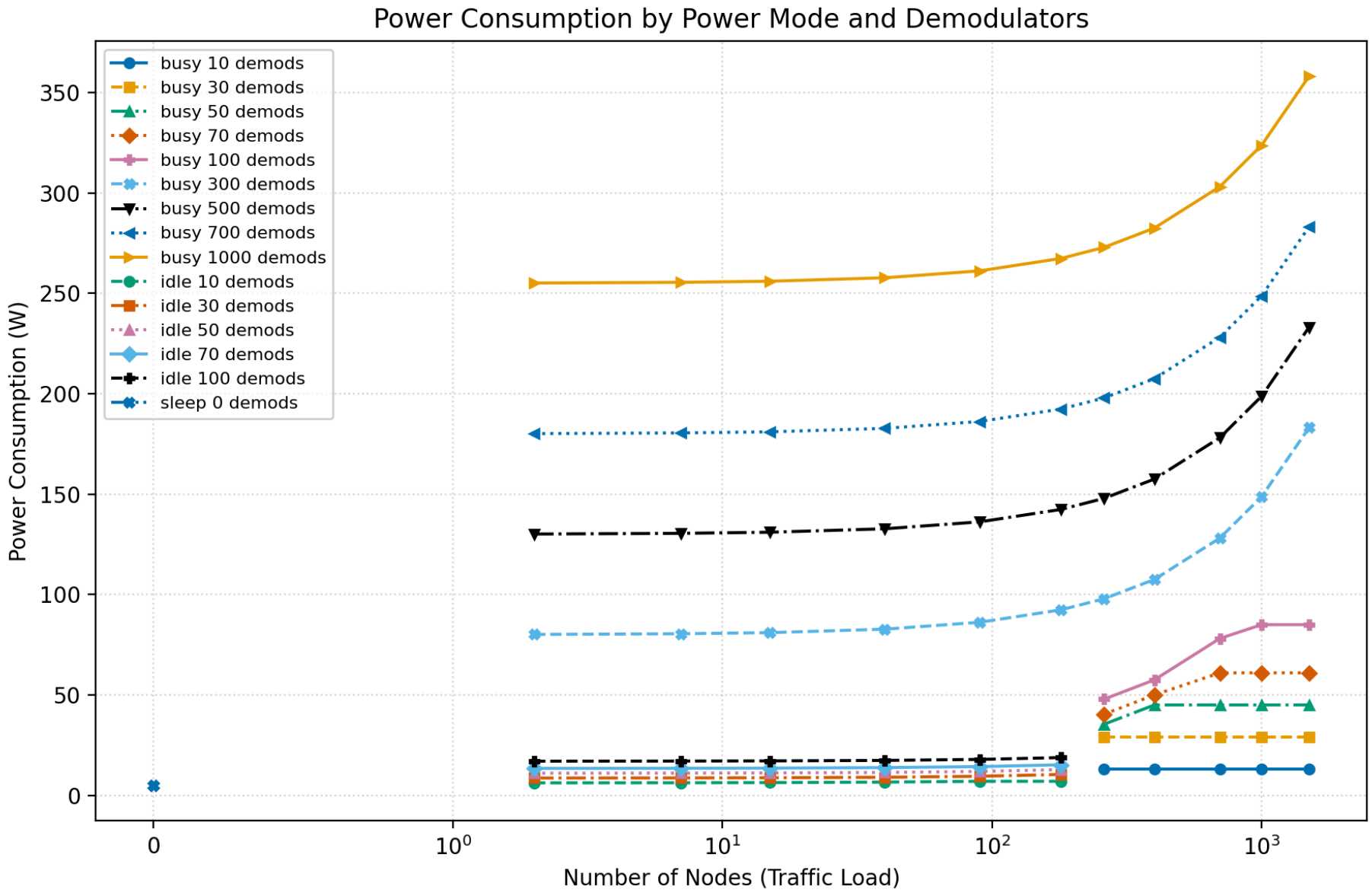
Mode-wise trend for the secondary KPI.



Results

Power Consumption by Mode

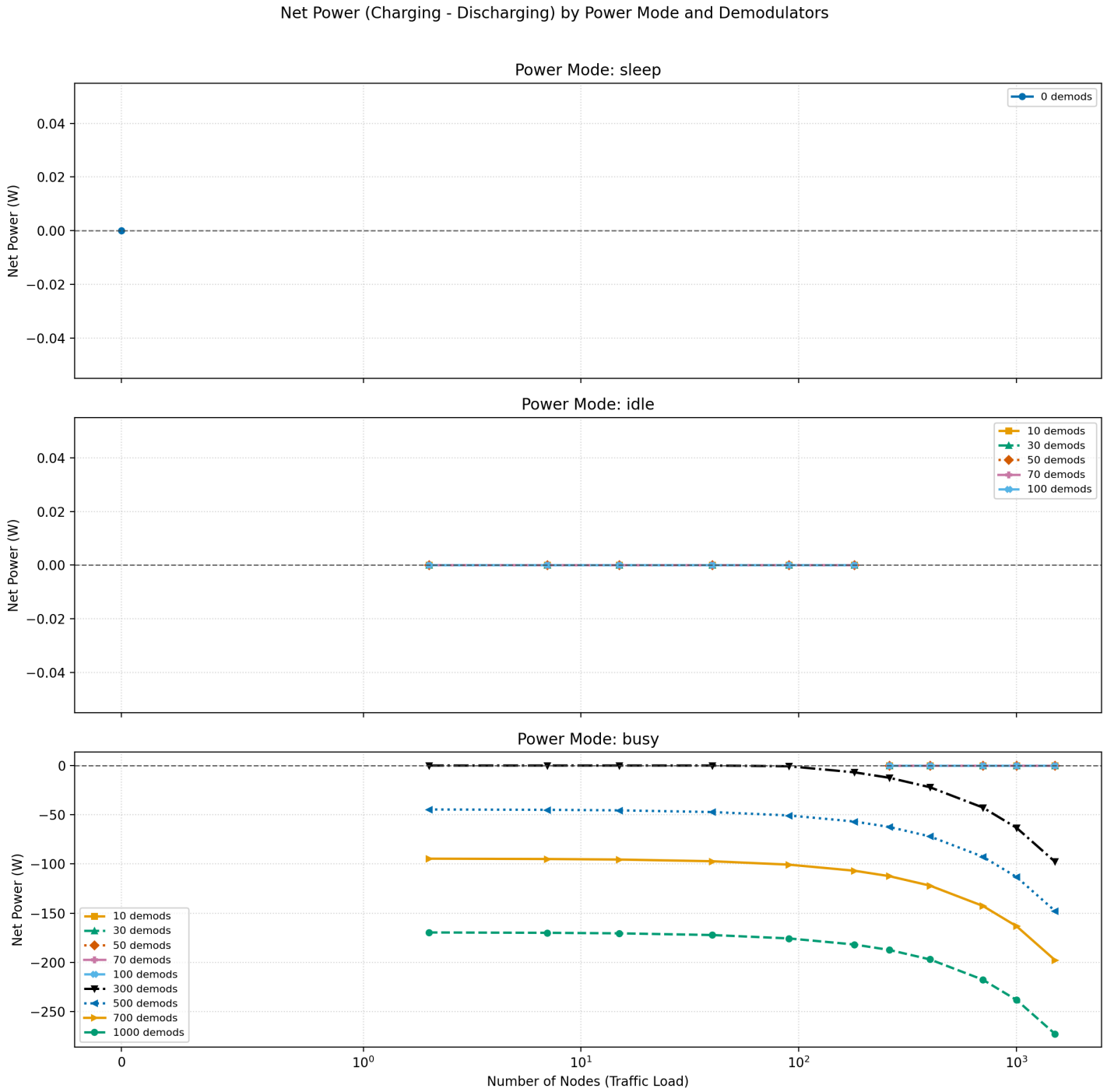
Receiver power trend across operating modes.



Results

Net Power by Mode

Charging minus discharging across traffic load.



Why These Equations

- SoC energy update (adapted from Coulomb-counting literature): 10.3390/en14144074
- Utilization-power linear model: 10.1145/1273440.1250665
- Energy proportionality: 10.1109/MC.2007.443

More Power-System References

- Spacecraft EPS balance model:
  - 10.1016/j.actaastro.2020.10.036
- CubeSat EPS survey:
  - https://link.springer.com/article/10.1007/s43937-025-00069-5
- CC-CV charge dynamics:
  - 10.1016/j.est.2020.101342